Aarogya Drishti: Project Documentation and Build Plan

AAROHAN Inter-Centre Hackathon 2026 | ITDA Paderu | 27 Oct to 4 Nov 2026 | Synthetic data only. NOT FOR CLINICAL USE.

1. Project at a glance

Item	Detail
Goal	One working platform from village oral screening to specialist diagnosis, treatment linkage and district reporting
Users	Health worker, Medical Officer (MO), Specialist, Navigator, District Admin
Team	Max 5 students, one team per student
Prize	Rs 10,000 per winning team (not per student)
Code freeze	4 Nov 2026, 10:00 AM IST
Data	Synthetic patients and licensed demo images only
Languages	Telugu and English with one-touch switch
Must run on	Real Android phone (Chrome PWA) and a computer, on a deployed HTTPS URL
Disqualifiers	Real patient data, presenting AI as clinical, exposed records, plagiarism, hardcoded secrets

2. Non-negotiable rules (build these into the design)

1. AI output is advisory and stored separately from the signed clinician conclusion.
2. The app never confirms or stages cancer by itself and never tells a patient they are cancer-free. Only a clinician-entered pathology result can mark a case as confirmed in the simulation.

3. Every mock or research model output carries the label NOT FOR CLINICAL USE, model version and timestamp.
4. The clinical workflow must keep working if the AI service is down.
5. Every API call is authenticated and authorized at patient/facility level, not just by role (deny by default).
6. No bounding boxes or heatmaps unless a real supervised localization model exists.
7. No secrets in Git. Only .env.example with placeholders.
8. Offline records are marked synced only after server acknowledgement.

3. Roles and permission matrix

Capability	Health Worker	Medical Officer	Specialist	Navigator	District Admin
Register patient, consent	Yes (own facility)	No	No	No	No
Screening intake, photos	Yes (assigned)	View	View referred only	No	No
Run AI analysis	Yes	View	View	No	No
Clinical review and sign	No	Yes	Consult notes only	No	No
Create referral	No	Yes	Yes (onward)	No	No
Book video/physical appointment	No	Yes	Confirm	Yes	No
Investigations and pathology result	No	Order/view	Yes	View status only	No
Enter final diagnosis	No	Yes	Yes	No	No
Navigation tasks	View assigned follow-up	View	No	Yes	No
View images	Own cases	Assigned cases	Referred cases	No (unless needed)	No
Dashboard	No	Facility summary	No	Own queue	Aggregates only
Audit log	No	No	No	No	Aggregate/system view

4. Architecture

Data flow: Health worker PWA, then authenticated FastAPI, then PostgreSQL plus private image storage, then AI adapter (research only), then stored AI observation, then Medical Officer, then video or physical specialist, then diagnostics, then navigator, then aggregate dashboard.

Stack

Layer	Choice
Frontend	React + TypeScript + Vite PWA (i18n with Telugu/English dictionary)
Backend	FastAPI, SQLAlchemy, Alembic, Pydantic
Database	PostgreSQL
Image storage	Private S3-compatible bucket (MinIO locally) with short-lived signed URLs
Image QC	OpenCV (blur, brightness, glare)
AI	Pluggable adapter: mock-v1 default, optional EfficientNet-B0 research model
Video	Approved SDK (for example Jitsi JaaS, Daily, LiveKit) with time-limited tokens
Offline	Service worker + IndexedDB queue with idempotency keys
Deploy	Docker Compose locally, managed container/VM plus managed Postgres online
Monitoring	Structured logs (no PHI), /healthz, uptime check

Folder structure

```
aarogya-drishti/
  apps/web/            # PWA: health worker, clinician, navigator, admin UI
  services/api/        # FastAPI, auth, workflow, migrations
  services/ai/         # quality checker + mock/research adapters
  infra/               # docker-compose, deployment manifests
  docs/                # PRD, architecture, ER, API, threat model, test
report
  tests/               # unit, integration, E2E, security
  .env.example
  README.md
```

5. Database schema

5.1 Table summary

Table	Purpose	Key rules
facilities	PHC, hospital, diagnostic centre	Mandal/PHC filters
users	Accounts and roles	Unique login, active flag, MFA for privileged
patients	Demographics and residence	UUID, duplicate hint, PII restricted
consents	Purpose-specific consent	Append-only, versioned
screenings	One screening visit	Clinician findings separate from AI
images	Oral photo metadata	Private key, checksum, consent gate
ai_analyses	Model outputs	Always clinical_use = false
clinical_reviews	MO signed review	Versioned, only clinician can finalize
referrals	Specialist referral	Explicit clinical decision
appointments	Video and physical visits	Status history, no public links
appointment_status_history	Status transitions	Append-only
investigations	Biopsy, pathology, imaging	Separate from AI label
diagnoses	Clinician-entered result	Only clinician roles
navigation_tasks	Follow-up work items	Overdue escalation
contact_attempts	Navigator calls/visits	Linked to task
idempotency_keys	Safe retries	Unique per user + key
audit_events	Who did what	Append-only, no PHI in metadata

Table	Purpose	Key rules
case_state_history	Case state transitions	Enforced transitions

5.2 Relationships

```
facilities 1-N users
facilities 1-N patients (registered_at)
patients 1-N consents | screenings
screenings 1-N images | clinical_reviews
images 1-N ai_analyses
clinical_reviews 1-N referrals
referrals 1-N appointments | investigations | navigation_tasks
investigations 1-N diagnoses
navigation_tasks 1-N contact_attempts
```

5.3 PostgreSQL DDL (core tables)

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;

CREATE TYPE user_role AS ENUM
('health_worker','medical_officer','specialist','navigator','district_admin')
;
CREATE TYPE case_state AS ENUM
('DRAFT','CONSENTED','SCREENED','AWAITING_CLINICIAN','REVIEWED','REFERRAL_PLA
NNED','APPOINTED','ASSESSED','DIAGNOSTICS_PENDING','RESULT_REVIEWED','CARE_LI
NKED','CLOSED','RESCHEDULED','MISSED','DECLINED','LOST_CONTACT');
CREATE TYPE appt_status AS ENUM
('REQUESTED','CONFIRMED','COMPLETED','MISSED','CANCELLED','RESCHEDULED');
CREATE TYPE appt_kind AS ENUM ('VIDEO','PHYSICAL');

CREATE TABLE facilities (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name TEXT NOT NULL,
  type TEXT NOT NULL,            -- PHC, HOSPITAL, DIAGNOSTIC
  mandal TEXT, district TEXT,
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
  email TEXT UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,   -- argon2/bcrypt
  full_name TEXT NOT NULL,
  role user_role NOT NULL,
  facility_id UUID REFERENCES facilities(id),
  mfa_secret TEXT,
  status TEXT NOT NULL DEFAULT 'active',
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE patients (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  patient_code TEXT UNIQUE NOT NULL,       -- app-generated ID
  full_name TEXT NOT NULL,
  dob DATE, age_years INT,
  sex TEXT,
  phone TEXT,
  village TEXT, habitation TEXT, gram_panchayat TEXT,
  mandal TEXT, phc_id UUID REFERENCES facilities(id),
  duplicate_hint TEXT,
  registered_by UUID REFERENCES users(id),
  state case_state NOT NULL DEFAULT 'DRAFT',
  created_at TIMESTAMPTZ DEFAULT now(),
  deleted_at TIMESTAMPTZ
);

CREATE TABLE consents (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  patient_id UUID NOT NULL REFERENCES patients(id),
  purpose TEXT NOT NULL,         -- EXAMINATION, IMAGING, TELECONSULT,
DATA_PROCESSING
  granted BOOLEAN NOT NULL,
  language TEXT NOT NULL,        -- te / en
  version TEXT NOT NULL,
  captured_by UUID REFERENCES users(id),
  captured_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE screenings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  patient_id UUID NOT NULL REFERENCES patients(id),
  examiner_id UUID NOT NULL REFERENCES users(id),
  screened_at TIMESTAMPTZ NOT NULL,
```

```
  tobacco_smoking BOOLEAN, tobacco_chewing BOOLEAN, gutkha BOOLEAN,
  areca_nut BOOLEAN, reverse_smoking BOOLEAN,
  habit_duration_months INT, habit_frequency TEXT,
  symptoms JSONB,                -- ulcer, white/red patch, pain, swallowing,
mouth opening, bleeding, growth
  symptom_duration_days INT,
  past_screening TEXT, referral_history TEXT,
  exam_findings TEXT,            -- health worker observation
  client_uuid UUID UNIQUE,       -- offline-created ID
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE images (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  screening_id UUID NOT NULL REFERENCES screenings(id),
  view TEXT NOT NULL,            -- CHEEK, TONGUE, PALATE, GUMS, FLOOR
  site_note TEXT,
  storage_key TEXT NOT NULL,     -- private, never public URL
  checksum_sha256 TEXT NOT NULL,
  quality_status TEXT NOT NULL,  -- acceptable / recapture
  quality_reason TEXT,
  uploaded_by UUID REFERENCES users(id),
  created_at TIMESTAMPTZ DEFAULT now(),
  UNIQUE (screening_id, checksum_sha256)
);

CREATE TABLE ai_analyses (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  image_id UUID NOT NULL REFERENCES images(id),
  provider TEXT NOT NULL,        -- mock / research
  model_version TEXT NOT NULL,
  research_label TEXT,           -- insufficient_image / suspicious_example /
none_identified
  score NUMERIC,                 -- nullable; not a probability
  status TEXT NOT NULL,          -- queued, done, failed
  clinical_use BOOLEAN NOT NULL DEFAULT false CHECK (clinical_use = false),
  warning TEXT NOT NULL DEFAULT 'NOT FOR CLINICAL USE',
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE clinical_reviews (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  screening_id UUID NOT NULL REFERENCES screenings(id),
```

```
  doctor_id UUID NOT NULL REFERENCES users(id),
  findings TEXT NOT NULL,
  ai_agreement TEXT,             -- AGREE / DISAGREE / DEFER
  discrepancy_note TEXT,
  decision TEXT NOT NULL,        -- ROUTINE_FOLLOWUP / SPECIALIST_REVIEW /
URGENT_REFERRAL / RECAPTURE
  version INT NOT NULL DEFAULT 1,
  signed_at TIMESTAMPTZ,
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE referrals (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  review_id UUID NOT NULL REFERENCES clinical_reviews(id),
  patient_id UUID NOT NULL REFERENCES patients(id),
  facility_id UUID REFERENCES facilities(id),
  specialty TEXT NOT NULL,
  priority TEXT NOT NULL,        -- ROUTINE / URGENT
  reason TEXT NOT NULL,
  status TEXT NOT NULL DEFAULT 'PLANNED',
  created_by UUID REFERENCES users(id),
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE appointments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  referral_id UUID NOT NULL REFERENCES referrals(id),
  kind appt_kind NOT NULL,
  clinician_id UUID REFERENCES users(id),
  facility_id UUID REFERENCES facilities(id),
  slot_start TIMESTAMPTZ NOT NULL,
  slot_end TIMESTAMPTZ NOT NULL,
  status appt_status NOT NULL DEFAULT 'REQUESTED',
  video_room_ref TEXT,           -- no public links; tokens issued per user,
short expiry
  attendance TEXT,               -- ATTENDED / NO_SHOW (separate from
outcome)
  clinician_outcome TEXT,
  missed_reason TEXT,
  created_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE appointment_status_history (
```

```
  id BIGSERIAL PRIMARY KEY,
  appointment_id UUID NOT NULL REFERENCES appointments(id),
  from_status appt_status, to_status appt_status NOT NULL,
  changed_by UUID REFERENCES users(id),
  changed_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE investigations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  referral_id UUID NOT NULL REFERENCES referrals(id),
  kind TEXT NOT NULL,            -- REEXAM, BIOPSY, HISTOPATHOLOGY, CT, MRI
  state TEXT NOT NULL,           -- ADVISED, SCHEDULED, COLLECTED,
REPORT_PENDING, RECEIVED, CLINICIAN_REVIEWED
  report_key TEXT,               -- private storage key
  ordered_by UUID REFERENCES users(id),
  reviewed_by UUID REFERENCES users(id),
  updated_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE diagnoses (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  investigation_id UUID NOT NULL REFERENCES investigations(id),
  category TEXT NOT NULL,        -- SCREEN_POSITIVE, CLINICAL_SUSPICION,
PRECANCER, PATHOLOGY_CONFIRMED_MALIGNANCY, BENIGN
  entered_by UUID NOT NULL REFERENCES users(id),
  note TEXT,
  treatment_linkage TEXT,
  signed_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE navigation_tasks (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  referral_id UUID NOT NULL REFERENCES referrals(id),
  owner_id UUID REFERENCES users(id),
  kind TEXT NOT NULL,            -- SPECIALIST_VISIT, INVESTIGATION,
TREATMENT_LINKAGE, MISSED_VISIT
  due_at TIMESTAMPTZ NOT NULL,
  status TEXT NOT NULL DEFAULT 'OPEN',   -- OPEN, IN_PROGRESS, ESCALATED,
CLOSED
  transport_support BOOLEAN DEFAULT false,
  created_at TIMESTAMPTZ DEFAULT now()
);
```

```
CREATE TABLE contact_attempts (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  task_id UUID NOT NULL REFERENCES navigation_tasks(id),
  method TEXT, outcome TEXT, notes TEXT,
  by_user UUID REFERENCES users(id),
  at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE idempotency_keys (
  user_id UUID NOT NULL REFERENCES users(id),
  key TEXT NOT NULL,
  request_hash TEXT NOT NULL,
  response_body JSONB,
  created_at TIMESTAMPTZ DEFAULT now(),
  PRIMARY KEY (user_id, key)
);

CREATE TABLE case_state_history (
  id BIGSERIAL PRIMARY KEY,
  patient_id UUID NOT NULL REFERENCES patients(id),
  from_state case_state, to_state case_state NOT NULL,
  changed_by UUID REFERENCES users(id),
  changed_at TIMESTAMPTZ DEFAULT now()
);

CREATE TABLE audit_events (
  id BIGSERIAL PRIMARY KEY,
  actor_id UUID, actor_role TEXT,
  event TEXT NOT NULL,           -- IMAGE_VIEW, REVIEW_SIGNED,
DIAGNOSIS_ENTERED, LOGIN, EXPORT ...
  resource_type TEXT, resource_id UUID,
  metadata JSONB,                -- IDs and codes only, never PHI
  at TIMESTAMPTZ DEFAULT now()
);

CREATE INDEX ON patients (phc_id, mandal);
CREATE INDEX ON screenings (patient_id, screened_at DESC);
CREATE INDEX ON navigation_tasks (status, due_at);
CREATE INDEX ON audit_events (resource_type, resource_id, at);
```

Make audit_events, consents, case_state_history and appointment_status_history append-only: revoke UPDATE/DELETE from the app database role, or add a trigger that raises an exception.

5.4 Case state machine

```
DRAFT > CONSENTED > SCREENED > AWAITING_CLINICIAN > REVIEWED >
REFERRAL_PLANNED
  > APPOINTED > ASSESSED > DIAGNOSTICS_PENDING > RESULT_REVIEWED >
CARE_LINKED | CLOSED
Tracked exceptions: RESCHEDULED, MISSED, DECLINED, LOST_CONTACT
```

Transition	Allowed actor	Guard
DRAFT to CONSENTED	Health worker	Imaging consent granted
CONSENTED to SCREENED	Health worker	Screening saved, at least one acceptable image or manual note
SCREENED to AWAITING_CLINICIAN	System	Submit for review
AWAITING_CLINICIAN to REVIEWED	Medical Officer	Review signed
REVIEWED to REFERRAL_PLANNED	Medical Officer	Decision is referral
REFERRAL_PLANNED to APPOINTED	MO / Navigator	Appointment CONFIRMED
APPOINTED to ASSESSED	Specialist	Appointment COMPLETED + outcome
ASSESSED to DIAGNOSTICS_PENDING	Specialist	Investigation ordered
DIAGNOSTICS_PENDING to RESULT_REVIEWED	Clinician	Report received and reviewed
RESULT_REVIEWED to CARE_LINKED / CLOSED	Clinician / Navigator	Diagnosis entered by clinician

No AI result may move a case forward by itself. Reject every transition not in this table with HTTP 409.

6. API specification

All routes need authentication, patient/facility authorization, validation, pagination for lists, standard error codes, and Idempotency-Key header on retryable writes.

Method + route	Purpose	Who
POST /auth/login	Session/JWT with expiry	All
POST /auth/mfa/verify	MFA for privileged users	MO, specialist, admin
GET /me	Current user, role, facility	All
POST /patients	Register patient (returns duplicate warnings)	Health worker
GET /patients/{id}	Longitudinal record	Assigned care team
POST /patients/{id}/consents	Purpose-specific consent	Health worker
POST /screenings	Create screening + history (accepts client_uuid)	Health worker
POST /screenings/{id}/images	Upload photo (needs consent)	Assigned health worker
GET /images/{id}/url	Short-lived signed URL, audited	Authorized clinicians
POST /ai/analyze	Queue non-clinical analysis	Screening user
GET /ai/analyses?image_id=	AI history	Care team
GET /queue	Role-scoped work queue	MO, specialist, navigator
POST /reviews	Save and sign clinical review	Medical Officer
POST /referrals	Create referral + printable slip	Medical Officer
GET /slots	Available appointment slots	MO, navigator
POST /appointments	Book / request	MO, navigator

Method + route	Purpose	Who
PATCH /appointments/{id}	Confirm, reschedule, cancel, attendance, outcome	Clinician / navigator
POST /appointments/{id}/video-token	Time-limited token for authorized participant	Participants only
POST /investigations	Order biopsy/pathology	Specialist, MO
PATCH /investigations/{id}	Update state, attach report	Specialist
POST /diagnoses	Clinician-entered result	MO, specialist
PATCH /navigation/{id}	Contact attempt, status	Navigator
GET /dashboard/summary	Aggregate counts and rates	District admin
GET /dashboard/export	Aggregate export only	District admin
GET /healthz	Health check	Public

Standard AI response (research demo)

```
{
  "analysis_id": "DEMO-AI-009",
  "screening_id": "DEMO-CASE-001",
  "image_quality": "acceptable",
  "provider": "mock",
  "model_version": "mock-v1",
  "research_label": "suspicious_example",
  "score": null,
  "clinical_use": false,
  "requires_clinician_review": true,
  "warning": "NOT FOR CLINICAL USE"
}
```

7. Module-by-module build guide

Module	What to build	How to build it	Done when
A. Registration and history	Patient form, exposure and symptom checklist, consent, duplicate check	Multi-step Telugu/English form; generate patient_code server-side; duplicate hint on name + age + village + phone; consent gates before screening	Patient and visit saved; duplicate warning shown
B. Smartphone screening	Guided camera, preview, retake, site tag, quality check	getUserMedia with overlay frame; client resize; upload to API; OpenCV check returns acceptable/recapture	Real phone captures 5 views; blurry image rejected
C. AI assessment	Adapter interface, mock model, stored outputs, warnings	Abstract class `analyze(image) -> dict`; mock returns fixed labels; timeouts and failure handling; version stored per row	Image to AI to persisted result; AI down does not block
D. MO and specialist review	Queue, case viewer with zoom, signed review, referral	Role-scoped queue query; review form with agree/disagree/defer; sign sets signed_at and locks version	Doctor overrides AI and referral created
E. Video	Slots, booking, token, join	Backend creates room, issues per-user JWT with 15-30 min expiry; recording off	Two accounts join; expired token denied
F. Physical appointment	Specialty, hospital, slot, printable slip, missed visit	Slip as HTML print or PDF with Telugu font embedded; missed status auto-creates navigation task	Slip prints in Telugu; missed flow works

Module	What to build	How to build it	Done when
G. Diagnostics	Investigation states, report upload, clinician diagnosis	State machine; report in private storage; diagnosis only by clinician roles	Simulated biopsy to pathology to diagnosis
H. Longitudinal record	Timeline view	Single query joining all events by patient; show actor, time, version	Full journey visible in one timeline
I. Navigation	Task queue, contact log, escalation	Daily job marks overdue and escalates; non-sensitive reminder text	Overdue shown and escalated
J. Dashboard	Aggregate cards, filters, export	SQL aggregates by village/PHC/mandal; no patient identifiers	Seeded 12 cases and 4 referrals show exact counts

8. Image quality and AI pipeline

Image checks (services/ai):

1. Validate extension and decode with Pillow/OpenCV; reject non-images (renamed PDF).
2. Enforce max size (for example 8 MB) and minimum dimensions.
3. Fix EXIF orientation, strip metadata, compute SHA-256.
4. Blur: variance of Laplacian below threshold means recapture.
5. Brightness: mean grayscale too low or too high means recapture.
6. Glare: fraction of saturated pixels above threshold means recapture.
7. Calibrate thresholds on your demo images and record them in docs.

AI adapter levels:

Level	What	When
1 (required)	Mock adapter with fixed demo labels	Day 4
2 (optional)	EfficientNet-B0 fine-tuned on the licensed dataset	Only if core system is complete

Level	What	When
3 (never)	Real clinical use	Not allowed

If you train the research model: write DATASET_CARD.md; remove corrupt files; split original source images into train/val/test before augmentation (no leakage); fine-tune on train only; report sensitivity, specificity, precision, F1, confusion matrix and sample sizes; write a MODEL_CARD.md with limitations. Output labels stay research-only.

9. Telugu and offline-first design

Area	Implementation
i18n	react-i18next with te.json and en.json; no hardcoded English; toggle always visible
Fonts	Noto Sans Telugu bundled locally; test on Android and in PDF
Usability	44px+ touch targets, icons with labels, progress steps, saved confirmation, few typed fields
Offline storage	IndexedDB drafts; document that browser storage is not fully protected; encrypt payloads with WebCrypto where feasible
Sync queue	Each write has client UUID + Idempotency-Key; states pending, uploading, confirmed, failed
Conflicts	Show resolution screen; never discard clinician notes
Jury test	Airplane mode, create patient and draft, close/reopen, reconnect, sync once, force retry, exactly one server record

10. Security checklist

Area	Requirement
Auth	Argon2/bcrypt, short session expiry, refresh handling, MFA for MO/specialist/admin, no shared accounts
Authorization	Role AND patient/facility check in every query; deny by default; tests for each endpoint
Transport and storage	HTTPS, encrypted DB and backups, private bucket, signed URLs under 5 minutes
Uploads	Decode, size limit, safe filenames, no path traversal, strip metadata
Audit	Log image views, exports, clinical edits, diagnosis, permission changes; no PHI in logs
Secrets	Env vars or secret manager; .env gitignored; rotate after event
Threats to cover	Stolen phone, expired/replayed token, IDOR, mass assignment, leaked signed URL, malicious upload, shared video link, backup leak
Governance	Synthetic data only; demo reset script; disable test users after event; mention DPDP and telemedicine compliance in roadmap

11. Test plan

Test	Input	Expected
Normal journey	Consent to screening to photo to review to video to biopsy	One continuous record
Blurred photo	Defocused image	Recapture warning
Invalid file	PDF renamed .jpg	Rejected after decode
Missing consent	Upload before consent	Rejected with explanation

Test	Input	Expected
Wrong patient	Doctor A opens unrelated case	403
AI down	Stop AI container	Case intact; manual review works
Network loss	Disconnect during upload	Retry, no duplicate image
Double submit	Same POST + same key	Single record
False-negative AI	AI low, clinician suspicious	Referral still possible
False-positive AI	AI suspicious, clinician disagrees	Discrepancy documented
Missed appointment	Mark missed	Navigation task created
Bad video token	Expired or wrong token	Join denied; reschedule possible
Long Telugu text	Switch language, PDF, mobile	No overflow
Concurrent edit	Two users edit one review	Conflict or version error
Dashboard	12 cases, 4 referrals seeded	Exact counts
Audit	Change pathology status	Actor/time/action stored
Invalid transition	Skip from SCREENED to CARE_LINKED	409
Oversized image	Over limit	413
Traversal filename	../../x.jpg	Sanitized or rejected

Required evidence: unit tests (pytest), API integration tests, Playwright mobile E2E, authorization tests, offline demo recording, known-issues list. Write failing security tests first, then fix.

12. Deployment and recovery

Step	Action
1	`docker compose up --build -d` (web, api, ai, postgres, minio)
2	Run Alembic migrations, then seed synthetic data (script: `make seed`)
3	Verify GET /healthz returns 200
4	Open HTTPS URL on Android; test camera, reload, offline
5	Restart API; confirm data persists
6	Confirm bucket is private and signed URLs expire
7	Run automated tests against the deployed URL
8	Take a DB backup and restore into a fresh environment
9	Provide `make reset` for demo data
10	After event: disable test users, rotate secrets

Free hosting tiers may sleep or wipe files. Use persistent database and object storage, and keep a local Docker route as backup.

13. Nine-day execution plan

Day	Date	Goal	Engineering lead	Frontend	AI / image	Clinical UI	QA / DevOps
1	27 Oct	Plan	Repo, PRD, schema draft	Wireframes, i18n keys	Risk register, dataset check	User journeys	Test strategy, CI setup
2	28 Oct	Foundation	FastAPI skeleton, migrations, login, RBAC	PWA shell, login, language toggle	Quality checker skeleton	Role layouts	Docker stack, healthz
3	29 Oct	Registration	Patient/consent/screening APIs	Registration, consent, checklist (Telugu)	Define adapter contract	Queue layout	Auth tests

Day	Date	Goal	Engineering lead	Frontend	AI / image	Clinical UI	QA / DevOps
4	30 Oct	Images and AI	Image upload, private storage, signed URLs	Camera, retake, preview	OpenCV QC + mock adapter	AI result panel	Upload negative tests
5	31 Oct	Review	Review, referral, state machine, audit	Offline draft queue	Stored AI history	MO queue, case viewer, sign	Authorization tests
6	1 Nov	Video and booking	Appointments, slots, token endpoint	Sync status UI	Optional research model	Booking, video join, slip	Two-device video test
7	2 Nov	Diagnostics and dashboard	Investigations, diagnoses, tasks, aggregates	Telugu polish	Model/dataset cards	Navigator, timeline, dashboard	Edge cases
8	3 Nov	Harden and deploy	Fix bugs, idempotency, backups	Real-phone testing	AI-down drill	Print Telugu PDF	Deploy, staging tests, test report
9	4 Nov	Demo	Freeze by 10:00 AM	Final checks	Final cards	Demo accounts	Backup route, video, rehearsal

Minimum end-to-end system should work by 31 Oct. Hold a 15-minute stand-up daily, merge twice a day, and keep a daily development log with commits.

14. Build steps (how to start)

```
git init aarogya-drishti && cd aarogya-drishti
mkdir -p apps/web services/api services/ai docs tests infra

# Frontend
cd apps && npm create vite@latest web -- --template react-ts
cd web && npm install && cd ../..

# Backend
cd services/api && python -m venv .venv
source .venv/bin/activate        # Windows: .venv\Scripts\activate
pip install fastapi 'uvicorn[standard]' sqlalchemy alembic \
  'psycopg[binary]' pydantic-settings python-multipart
pip freeze > requirements.txt     # pin and review versions
```

Build order (vertical slices, each with passing tests before moving on):

1. Docker stack + health check + migrations + seed data

2. Auth, roles, facility scoping, audit middleware
3. Patient, consent, screening APIs and Telugu/English forms
4. Image upload, quality checks, mock AI, stored results
5. Review queue, signed review, referral
6. Appointments, video tokens, physical slip
7. Investigations, diagnosis, navigation tasks
8. Timeline and dashboard aggregates
9. Offline sync, security tests, deployment, documentation

Tips for using Antigravity or other AI assistants: ask for a design first, inspect all generated code, run the tests yourself, commit only accepted changes, never paste real records or secrets, and disclose AI tool use in your submission.

15. Final live demo script (10 minutes)

1. Open the deployed app in Telugu on a phone.
2. Register synthetic patient: 46-year-old from G. Madugula, tobacco chewing, persistent white cheek patch.
3. Record consent and symptoms.
4. Upload a good demo image, then a blurry one (recapture shown).
5. Run the mock AI; stop the AI service and show graceful recovery.
6. Log in as Medical Officer, review, disagree with AI if relevant, sign the decision.
7. Book a video consultation and join with a second account.
8. Create a physical referral, mark attendance, upload simulated pathology.
9. Specialist enters the final simulated diagnosis.
10. Navigator task updated; dashboard counts correct.
11. Show unauthorized access blocked, audit trail, and offline sync.
12. Explain limitations and the roadmap to an approved clinical pilot.

16. Submission checklist

Deliverable	Status
Git repo with daily commits and contribution record	
Deployed HTTPS demo URL + evaluator test accounts	
README (install, architecture, stack, credentials, known limits)	
Architecture diagram with trust boundaries; ER diagram	
Role-access matrix	
OpenAPI docs with examples	
Environment variable table (no secrets)	
AI model card, dataset card, validation and limitations	
Threat model	
Test report with failure cases and offline tests	
Deployment, backup and rollback guide	
Synthetic dataset + reset command	
Telugu UI demonstration	
5-minute demo video	
Disclosure of AI tools, templates, third-party code	
Roadmap from demo to approved clinical pilot	

17. Scoring map (100 marks)

Criterion	Marks	What earns points
End-to-end workflow and integration	20	Full journey works with one continuous record
AI engineering, image quality, safe integration	15	Quality checks, versioned adapter, clear non-clinical labels, AI-down resilience
MO review, video, appointments	15	Signed review, working two-party call, status tracking
Records, diagnostics, navigation	15	Timeline, investigation states, navigator follow-up
Security, privacy, access	15	Patient-level authorization, audit, private images, MFA
Telugu and offline reliability	10	Full Telugu flow, offline sync with no duplicates
Deployment, testing, documentation	10	Live URL, tests, reset script, complete docs

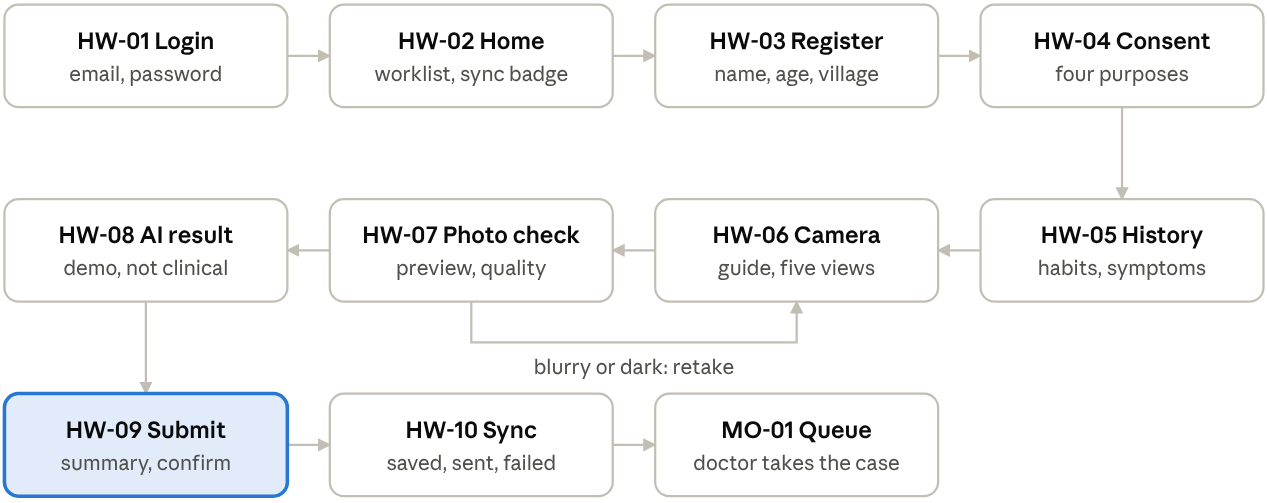
Tie-breakers: patient workflow, then security/privacy, then AI safety/integration. A safety violation can disqualify a team regardless of technical score.

18. Screen flows and page designs

Every screen has an ID. Health worker screens (HW) are mobile first. Medical Officer, specialist, navigator and admin screens (MO, SP, NV, AD) are designed for a laptop or tablet and must still work on a phone. Each table row says what is on the page and where the main button goes next.

18.1 Health worker app (mobile, Telugu and English)

Health worker records a case in ten screens, then the doctor takes over



health worker screen flow · 10 screens, 1 retake loop

The worker moves forward with one big Next button per page. A blurry photo loops back to the camera, and the last screens hand the case to the doctor queue.

Screen	Page design (top to bottom)	Click	Goes to	Rules
HW-01 Login	Logo, TE/EN toggle, email field, password field, full-width Sign in button	Sign in	HW-02 (MFA code page first for privileged roles)	Telugu error for wrong password; rate-limit attempts
HW-02 Home	Top bar with facility name, sync badge and language toggle; three big tiles (New patient, Today's cases, Pending sync); recent cases list with status chips	New patient	HW-03	Offline drafts show an orange Draft chip

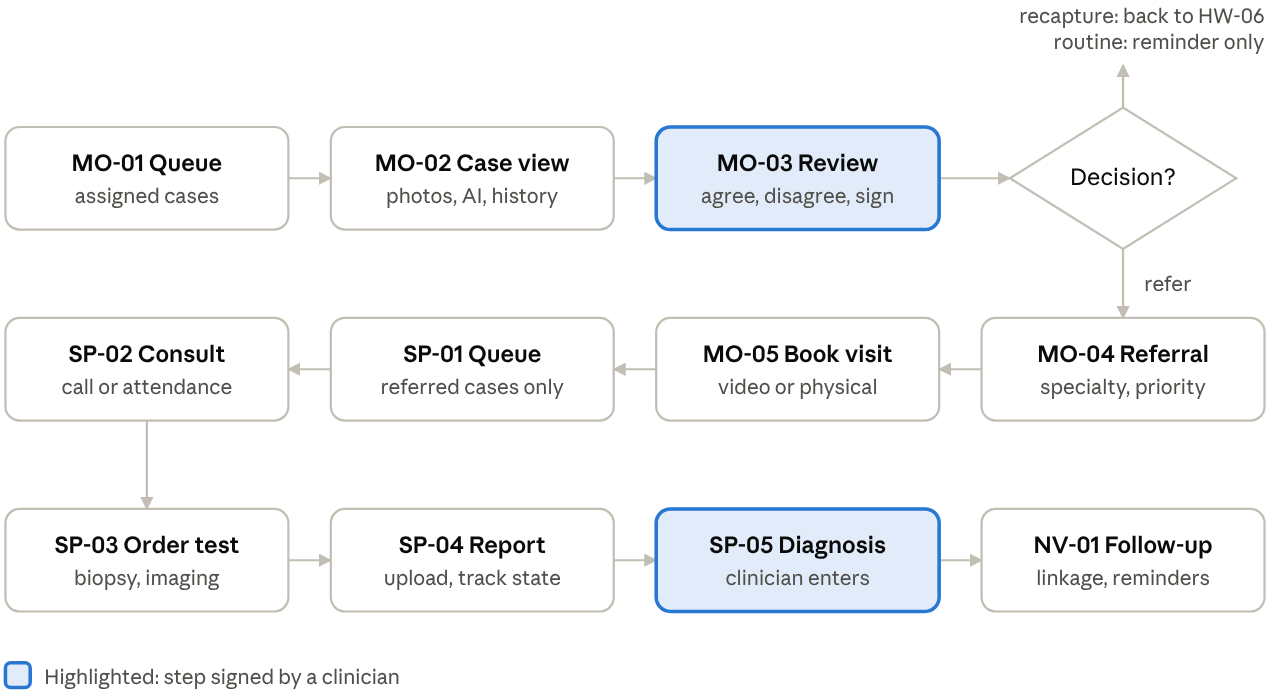
Screen	Page design (top to bottom)	Click	Goes to	Rules
HW-03 Register	Progress 1 of 5; name, age or date of birth, sex (3 big buttons), phone, village, habitation, Gram Panchayat, mandal, PHC (auto-filled from facility)	Next	HW-04; if a duplicate is found, a card offers Open existing record or Continue as new	Name and age required; server creates patient_code
HW-04 Consent	Progress 2 of 5; four toggle cards (examination, imaging, teleconsultation, data processing), each with an icon and one Telugu line; optional audio play	Next	HW-05	Next enabled only with examination and imaging consent; refusing imaging skips photos and uses a manual note
HW-05 History	Progress 3 of 5; habit rows (smoking, chewing, gutkha, areca nut, reverse smoking) with big Yes/No, duration and frequency pickers; symptom chips (ulcer, white patch, red patch, pain, swallowing, mouth opening, bleeding, growth); symptom duration; past screening	Next	HW-06	Minimal typing; chips and pickers only

Screen	Page design (top to bottom)	Click	Goes to	Rules
HW-06 Camera	Progress 4 of 5; live camera with a framing guide, view selector (cheek, tongue, palate, gums, floor of mouth), lighting tip text, large shutter button	Capture	HW-07	Camera permission denied shows Upload from gallery (demo images)
HW-07 Photo check	Preview with pinch zoom; quality badge (OK, blurry, dark, glare); body-site note; Retake and Use photo buttons	Use photo	Next view on HW-06, or HW-08 when all views are done	Retake goes back to HW-06; a failed check needs a note to continue
HW-08 AI result	Amber card: Demonstration AI, NOT FOR CLINICAL USE; label (image insufficient, suspicious for clinical review, none identified by demo model); model version; no score, no overlay	Next	HW-09	Never says healthy or cancer; if AI is down a grey card says the doctor will review
HW-09 Submit	Progress 5 of 5; summary card (patient, habits, symptoms, photo thumbnails, consent ticks) with Edit links; big Submit for doctor review button	Submit	HW-10	Sends client UUID and Idempotency-Key; state becomes AWAITING_CLINICIAN

Screen	Page design (top to bottom)	Click	Goes to	Rules
HW-10 Sync	Large tick and Saved message; status chip (pending, uploading, confirmed, failed); case ID; buttons New patient and Home	Home	HW-02	Marked confirmed only after the server acknowledges

18.2 Medical Officer and specialist portal (laptop or tablet)

Only clinicians sign the review and the final diagnosis



clinical screen flow · 12 screens, 1 decision

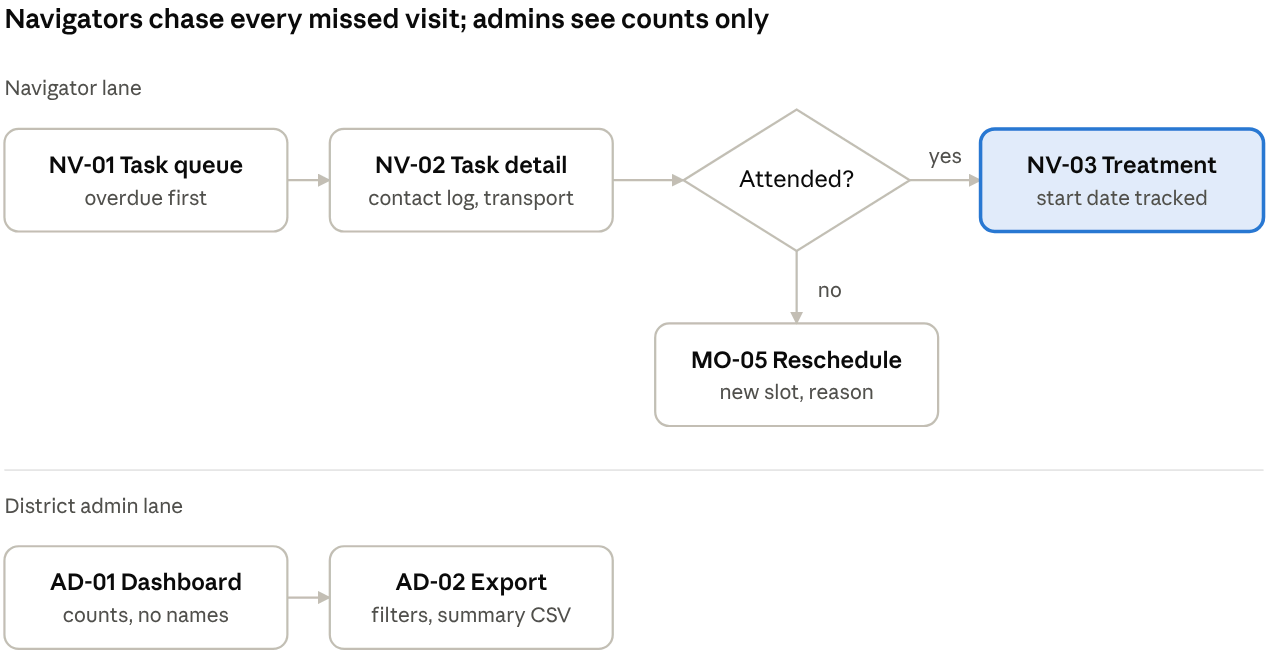
The Medical Officer decides after reading the case; only a referral continues to the specialist, who orders tests, records the result and enters the final diagnosis.

Screen	Page design (top to bottom)	Click	Goes to	Rules
MO-01 Queue	Left filter column (urgency, facility, date); table of cases with code, age and sex, village, advisory AI chip, waiting time, status; oldest first	Click a row	MO-02	Shows only cases of the doctor's facility
MO-02 Case view	Header with code, age, sex, village; tabs Summary (habits, symptoms), Photos (thumbnails, zoom, side by side), AI (versioned history, NOT FOR CLINICAL USE banner), Timeline; right panel with actions	Start review	MO-03	Every image view is written to the audit log; Request recapture sends the case back to the worker
MO-03 Review	Findings text box; AI agreement choice (agree, disagree, defer) plus discrepancy note; decision choice (routine follow-up, specialist review, urgent referral, recapture); Sign button with confirm dialog	Sign and save	MO-04 for referral; HW-06 for recapture; NV-01 reminder for routine	Signed review is locked; changes create a new version

Screen	Page design (top to bottom)	Click	Goes to	Rules
MO-04 Referral	Specialty and hospital dropdowns, priority (routine or urgent), reason pre-filled from the review; slip preview in Telugu and English	Create referral	MO-05	Reason is mandatory
MO-05 Book visit	Switch Video or Physical; calendar with slot chips; chosen clinician or hospital; confirmation summary	Confirm	Success page with printable slip, then MO-01	Status starts as REQUESTED, then CONFIRMED; reminder text hides clinical details
SP-01 Queue	Tabs Today, Upcoming, Results pending; cards for referred cases only	Open	SP-02	Specialist cannot browse other patients
SP-02 Consult	Case sidebar; for video: Join call button (time-limited token), camera preview, notes pane; for physical: attended or missed buttons with reason, clinician notes, outcome	Save outcome	SP-03 if tests are needed; NV-01 if missed; otherwise case closes	Attendance stored separately from clinical outcome; recording off
SP-03 Order test	Test type (re-examination, biopsy, histopathology, CT, MRI), facility, notes	Order	SP-04	State starts ADVISED

Screen	Page design (top to bottom)	Click	Goes to	Rules
SP-04 Report	Stepper: advised, scheduled, collected, report pending, received, reviewed; report upload (type and size checked)	Mark reviewed	SP-05	Every state change is audited
SP-05 Diagnosis	Category choice (screen-positive, clinical suspicion, precancer, pathology-confirmed malignancy, benign); note; treatment linkage; Sign button	Sign	NV-01	Only clinician roles; AI can never set this field
TL-01 Timeline	Vertical list of events: registration, consents, screenings, images, AI outputs, signed notes, calls, referrals, tests, diagnosis, follow-up; each with actor, time, version	Open from any case	Back to the case	Read-only; access follows care relationship

18.3 Navigator and district admin



navigator and admin flow · 5 screens, 1 decision

A missed visit sends the navigator back to rescheduling; the admin lane never opens a patient page.

Screen	Page design (top to bottom)	Click	Goes to	Rules
NV-01 Task queue	Tabs Overdue, Due today, Upcoming; one card per task with patient code, task type, due date, Call button	Open task	NV-02	Phone number shown only on the task; no photos
NV-02 Task detail	Contact log; buttons Called, Visited, No answer; missed-reason chips; transport support checkbox; next follow-up date	Save	NV-03 if attended; MO-05 if not attended	Each contact attempt is stored with time and user; reminder text has no clinical details

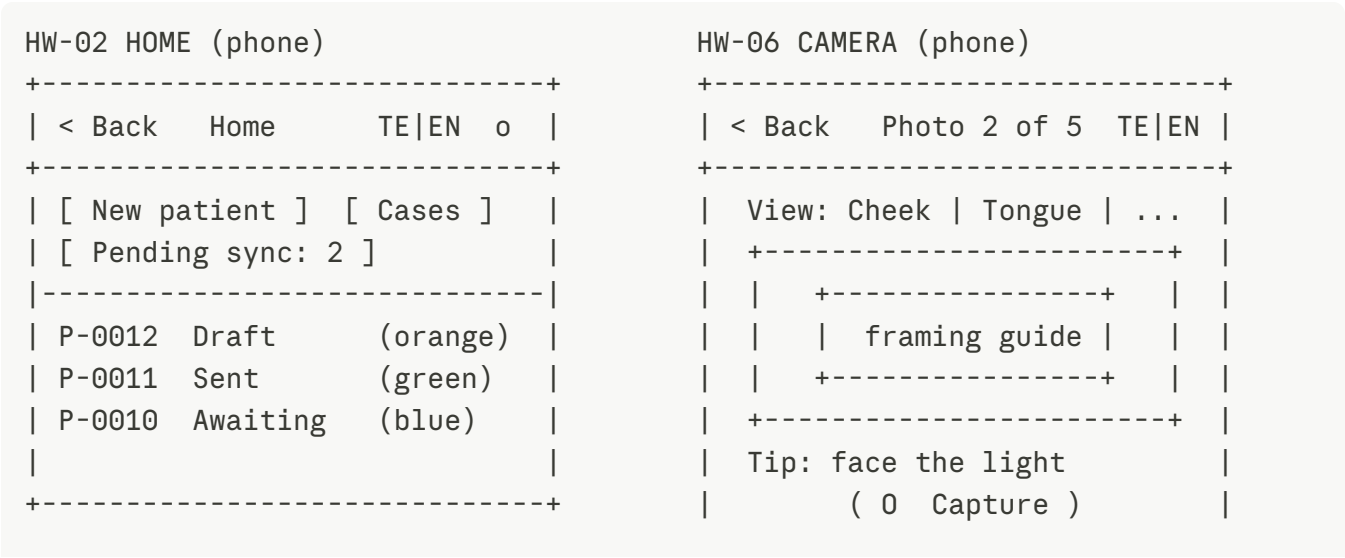
Screen	Page design (top to bottom)	Click	Goes to	Rules
NV-03 Treatment	Treatment facility, start date, status (started, pending, declined, lost contact); Close task button	Close	NV-01	Closure needs a clinician-entered diagnosis on the case
AD-01 Dashboard	Filter bar (mandal, PHC, village, date range); KPI cards for screenings, doctor reviews, suspicious findings, referrals, completed appointments, overdue cases, biopsy pending and completed, confirmed precancer or cancer, treatment linkage, lost to follow-up, average time to review and diagnosis; two simple charts	Export	AD-02	Aggregates only; no names, phone numbers or images
AD-02 Export	Chosen filters, preview of the summary table, Download CSV button	Download	Back to AD-01	Export is logged; no row-level patient data

18.4 Design rules for every page

Area	Rule
Mobile layout (HW)	Top bar (back, title, TE/EN toggle, sync badge); progress dots; one content card; sticky bottom bar with one full-width Next button at least 48px tall

Area	Rule
Desktop layout (MO, SP, NV, AD)	Left menu, top bar with role and facility, main area with table or case view, right action panel
Touch and type	Targets 44px or larger; body text 16px or larger; icon plus label on every button
Language	Every string comes from the te/en dictionary; toggle visible on every page; Telugu error messages
Status colours	Grey draft, blue pending, green confirmed or signed, amber warning or AI advisory, red failed or urgent. Never colour a patient as cancer or healthy
AI card	Always amber with the text NOT FOR CLINICAL USE, model version and timestamp; no score, no heatmap, no box
Clinician-signed items	Green badge with doctor name, time and version
Feedback	Visible Saved confirmation after every write; clear retry on failure; offline banner when disconnected
Empty and error states	Plain Telugu/English sentence plus one action button
Privacy	No patient names in browser titles, notifications or URLs; session timeout screen

18.5 Page wireframes (key screens)



```
                                          +-----------------------------+

HW-08 AI RESULT (phone)                   MO-02 CASE VIEW (laptop)
+-----------------------------+           +-----------------------------------
----------+
| < Back   AI result   TE|EN   |           | P-0012  46 M  G. Madugula
[Start review] |
+-----------------------------+           +-----------------------------------
----------+
| +-------------------------+ |           | Summary | Photos | AI | Timeline
|
| | DEMONSTRATION AI        | |           |-----------------------------------
----------|
| | NOT FOR CLINICAL USE    | |           | Photos (zoom)        | Habits:
chewing 8 yrs  |
| | Label: suspicious for   | |           | [img][img][img]      | Symptom:
white patch   |
| | clinical review         | |           |                      | AI
(advisory, amber)   |
| | Model: mock-v1          | |           |                      | [Request
recapture]    |
| +-------------------------+ |           +-----------------------------------
----------+
| [ Next ]                    |
+-----------------------------+

AD-01 DASHBOARD (laptop)
+-------------------------------------------------------------+
| Mandal [v]  PHC [v]  Village [v]  Dates [v]          [Export] |
+-------------------------------------------------------------+
| Screenings 12 | Reviews 10 | Suspicious 5 | Referrals 4        |
| Completed appts 3 | Overdue 1 | Biopsy pending 2 | Linked 1    |
+-------------------------------------------------------------+
| [ Screenings by week ]        [ Referral funnel ]            |
+-------------------------------------------------------------+
```

18.6 Build recipe for each page

Build each page in this order: route and empty page, translated text, form or list, API call, validation and error states, offline behaviour (HW pages), then the test in the last column.

Screen	Route	Main API call	Test before moving on
HW-01	/login	POST /auth/login	Wrong password, expired token
HW-02	/home	GET /me, GET /queue	Offline drafts appear in the list
HW-03	/patients/new	POST /patients	Duplicate warning appears
HW-04	/patients/:id/consent	POST /patients/{id}/consents	Image upload blocked without consent
HW-05	/screenings/new	Kept in local state, saved with HW-09	Long Telugu text does not overflow
HW-06	/screenings/:id/camera	None (local capture)	Works on a real Android phone
HW-07	/screenings/:id/photo	POST /screenings/{id}/images	Blurry image rejected; retry makes no duplicate
HW-08	/screenings/:id/ai	POST /ai/analyze	AI service stopped shows the grey fallback card
HW-09	/screenings/:id/submit	POST /screenings with Idempotency-Key	Double tap creates one record
HW-10	/done	Sync queue status	Airplane-mode drill ends with exactly one server record
MO-01	/queue	GET /queue	Doctor A cannot see another facility's cases

Screen	Route	Main API call	Test before moving on
MO-02	/cases/:id	GET /patients/{id}, GET /images/{id}/url	Image view appears in the audit log
MO-03	/cases/:id/review	POST /reviews	Signed review cannot be edited silently
MO-04	/cases/:id/referral	POST /referrals	Missing reason rejected
MO-05	/referrals/:id/book	GET /slots, POST /appointments	Invalid status change returns 409
SP-01	/specialist/queue	GET /queue	Specialist sees referred cases only
SP-02	/appointments/:id	PATCH /appointments/{id}, POST /appointments/{id}/video-token	Two accounts join; expired token denied
SP-03	/investigations/new	POST /investigations	Only clinician roles allowed
SP-04	/investigations/:id	PATCH /investigations/{id}	State order enforced; upload type checked
SP-05	/investigations/:id/diagnosis	POST /diagnoses	AI or worker role cannot set a diagnosis
TL-01	/patients/:id/timeline	GET /patients/{id}	Timeline shows actor, time and version
NV-01	/tasks	GET /queue	Overdue tasks appear first

Screen	Route	Main API call	Test before moving on
NV-02	/tasks/:id	PATCH /navigation/{id}	Missed visit creates reschedule path
NV-03	/tasks/:id/treatment	PATCH /navigation/{id}	Cannot close without clinician diagnosis
AD-01	/dashboard	GET /dashboard/summary	Seeded 12 cases and 4 referrals give exact counts
AD-02	/dashboard/export	GET /dashboard/export	Export contains no patient identifiers